

Derivation of the radial distribution for a 2D Gaussian (Rice / Rayleigh)

2D Gaussian with mean vector magnitude ν and standard deviation σ

Let X and Y be independent Gaussian random variables with common variance σ^2 and a mean vector with components μ_x, μ_y . Denote the magnitude of the mean vector by $\nu = \sqrt{\mu_x^2 + \mu_y^2}$.

The joint probability density is $f_{X,Y}(x,y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-\mu_x)^2 + (y-\mu_y)^2}{2\sigma^2}\right)$.

Switch to polar coordinates: $x = r\cos\theta$, $y = r\sin\theta$ with $r \geq 0$ and $\theta \in [0, 2\pi)$. The Jacobian is $dx\,dy = r\,dr\,d\theta$, so the joint density becomes

$f_{R,\Theta}(r,\theta) = \frac{r}{2\pi\sigma^2} \exp\left(-\frac{r^2 + \nu^2 - 2r\nu\cos(\theta - \phi)}{2\sigma^2}\right)$, where ϕ is the angle of the mean vector.

Integrate out θ to obtain the marginal radial density: $p_R(r) = \int_0^{2\pi} f_{R,\Theta}(r,\theta)\,d\theta = \frac{r}{2\pi\sigma^2} e^{-\frac{r^2 + \nu^2}{2\sigma^2}} \int_0^{2\pi} e^{\frac{r\nu}{\sigma^2}\cos(\theta - \phi)}\,d\theta$.

Use the modified Bessel function identity $I_0(z) = \frac{1}{2\pi} \int_0^{2\pi} e^{z\cos t}\,dt$. Thus one obtains the Rice distribution: $p_R(r) = \frac{r}{\sigma^2} \exp\left(-\frac{r^2 + \nu^2}{2\sigma^2}\right) I_0\left(\frac{r\nu}{\sigma^2}\right)$, $r \geq 0$.

In the special case $\nu = 0$ (zero-mean), $I_0(0) = 1$ and the radial density reduces to the Rayleigh distribution $p_R(r) = \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right)$.

Remarks: the factor r arises from the Jacobian of the polar transform and accounts for the circumference at radius r . For $\nu = 0$ the density vanishes at $r = 0$ and peaks at $r = \sigma$, while for $\nu > 0$ the Bessel factor shifts and reshapes the distribution.